



Learner-Stage-Aware AI Tutor Improves Learning Processes: Initial Evidence from a Field Experiment

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Abstract. Acquisition of cognitive skills depends on building problem-solving schemas. These schemas develop most effectively when instructional support aligns with learners' current level of expertise. Novices benefit from worked-out examples that make the solution rationale explicit, whereas more advanced learners benefit from prompts that elicit self-explanations, allowing them to complete solution steps on their own. Tutors based on Large Language Model (LLM) provide scalable, dialog-based assistance that can deliver such instructional support. However, there is scarce evidence on how personalizing tutoring based on initial skill stage with LLMs affects learning processes.

In this study, we investigate the effects of personalizing instructional support by stage-aware scaffolding in LLM-based tutors on learners' tutor usage, reasoning behavior, and mental effort. To this end, we developed an LLM-based tutor grounded in the cognitive skill acquisition theory. It uses worked-out examples with scaffolds for novices and fades this support out for more advanced learners. We used a randomized controlled experiment in a master's-level programming course to compare the stage-aware tutor with a baseline tutor. Learners were randomly assigned to one of the two tutor groups. Results indicate that learners using the skill-stage-aware tutor (vs. baseline) sent more messages and were active across more days, and their interactions were more likely to reflect deep rather than shallow engagement. Furthermore, learners who used the skill-stage-aware tutor invested higher mental effort in their schema construction. The results indicate that theory-grounded personalization to learners' initial skill stages can provide process-level scaffolding during self-study and shape learner-tutor dialogue toward cognitively productive activity.

Keywords: Early Skill Acquisition · Instructional Scaffolding · AI Tutoring Systems · Learner-AI Interaction

1 Introduction

Initial cognitive skill acquisition in well-structured domains, such as mathematics or computer programming, is fundamentally tied to the construction of abstract problem-solving schemas [23]. During this formative period, learners progress through a sequence of acquisition stages, ranging from initial conceptual understanding to example-based learning, early problem-solving, and eventually extensive practice [22]. Successful progression requires that learning activities align with the learner’s current stage and that learners accurately judge when they are ready to advance [31]. When stage and learning activity are misaligned, for example, when learners engage in problem-solving before having developed sufficient conceptual understanding, learning becomes compromised. Learners tend to rely on shallow strategies such as pattern matching, which in turn creates illusions of understanding [9]. These mismatches eventually lead to frustration, reduced motivation, and fragile knowledge structures that fail to support further learning [16, 28].

Rapid rise of LLM-based tutoring systems has amplified this longstanding instructional challenge. General-purpose LLMs such as ChatGPT provide immediate, high-quality solutions and explanations on demand, substantially lowering the barrier to advanced problem-solving. While this accessibility can be beneficial, it also increases the likelihood that learners prematurely shift to activities that exceed their current level of expertise, bypassing critical intermediate stages of schema construction. As a result, LLM-based tutoring risks systematically reinforcing stage–activity mismatches at scale, particularly in early phases of skill acquisition.

One promising response to this challenge is learner-stage-aware LLM-based tutoring, which we define here as tutoring where instructional support is explicitly adapted to the learner’s initial stage of skill acquisition.¹ Rather than providing uniform responses, a stage-aware tutor modulates its behavior by offering worked examples to novices, prompting self-explanations for intermediate learners, and gradually fading support for more advanced learners. While prior work has demonstrated the effectiveness of scaffolding and adaptive instruction in both human tutoring [3, 25] and Intelligent Tutoring System (ITSs) [13, 15, 16], there remains limited evidence on how learner-stage-aware personalization in LLM-based tutors shapes learning processes in authentic classroom settings.

In this paper, we design and evaluate an LLM-based AI tutor that provides learner-stage-aware scaffolding grounded in cognitive skill acquisition theory. We make three contributions. First, we operationalize established theories of early skill acquisition into a concrete framework for customizing LLM behavior. Second, we implement a randomized controlled trial in an authentic university classroom, comparing a learner-stage-aware AI tutor to a baseline AI tutor without stage-aware scaffolding, and documenting student adoption and usage of both systems under real instructional conditions. Third, we provide empirical evidence on process-level effects of learner-stage-aware tutoring, showing how skill-

¹ As opposed to real-time mastery tracking.

stage-aware scaffolding shapes learners' engagement patterns, reasoning-oriented dialogue, and mental effort.

2 Background and Research Questions

2.1 Cognitive Skill Acquisition

Cognitive skill acquisition is a process through which learners transform initially explicit knowledge into stable, efficient problem-solving competence. Prominent theoretical models [2, 29] describe this process as unfolding across three learning phases: an *early* phase, in which learners acquire a basic conceptual understanding of domain principles; an *intermediate* phase, in which they learn how these principles are applied to solve concrete problems; and a *late* phase, in which extensive independent practice leads to increased accuracy and automation.

Effective learning depends on engaging in the right learning activity at the right time. In particular, late-phase activities (i.e., independent problem-solving) are most beneficial only once learners are sufficiently prepared. Delaying the transition to this phase can lead to stagnation, where learners remain dependent on explicit guidance and fail to develop fluency [25]. Conversely, transitioning too early can overload learners' cognitive resources, resulting in superficial performance and fragile knowledge that does not transfer [16, 28]. Consequently, learners benefit most when instructional activities are closely aligned with their current learning phase.

2.2 Scaffolding Learners with Worked Examples and Fading

To support the early and intermediate phases of cognitive skill acquisition, several instructional paradigms have been proposed; for an overview, see [22]. A central principle underlying these approaches is *instructional scaffolding*, defined as the temporary provision of structured support that guides learners' reasoning and is gradually reduced as competence increases. One of the most influential and widely adopted implementations of scaffolding is the *worked-examples approach*, grounded in cognitive load theory and the worked-example effect. Within this framework, instruction is typically organized into three stages that align with the early and intermediate phases of cognitive skill acquisition.

In Stage 1 (corresponding to the early phase of skill acquisition), instructors provide fully worked solutions to reduce learners' extraneous cognitive load and free cognitive resources for schema construction [28]. In Stage 2 (corresponding to the early-intermediate phase of skill acquisition), learners are questioned on underlying principles and are exposed to multiple worked examples to develop an understanding of how these principles are applied in concrete problem-solving contexts. In Stage 3 (corresponding to the late-intermediate phase of skill acquisition), instructional support is gradually reduced, and learners are prompted to solve problems themselves, fostering deeper processing and consolidation of concepts. Stages 2 and 3 thus operationalize the principle of *fading*, whereby instructional guidance is progressively withdrawn as learners gain expertise [24].

In this paper, we focus on stages 1, 2 and 3 and refer to them as novice, early-intermediate and late-intermediate accordingly.

2.3 Scaffolding in Tutoring Systems

Research has consistently demonstrated the effectiveness of scaffolding in instruction: both with human tutoring [3, 25] and with ITSs [13, 15, 16]. ITSs are particularly well suited for educational settings because they provide scalable and standardized instructional support while enabling learning outcomes that can approach those achieved through human tutoring [5].

Recent work explores different ways of operationalizing scaffolding in ITSs. Kazemitabaar et al. introduce cognitive engagement techniques that scaffold problem solving by structuring when and how AI-generated solutions are revealed [13]. Wang et al. explore design choices for supporting novices' use of examples in creative, open-ended programming [32]. Akhuseyinoglu et al. implement interactive worked examples that scaffold reasoning through step-by-step explanations [1].

Moving to LLM-based tutoring, Ma et al. present a system that scaffolds algorithmic problem solving through learner-LLM co-decomposition, helping learners break down problems into structured steps while providing progressive hints [15]. Bhaskar et al. implement scaffolding through answer-aware prompt constraints that generate progressively specific hints [6], while Xie et al. employ guided-questioning strategies that scaffold learners' reasoning before providing solutions [33]. Bastani et al. show the effectiveness of LLM-based tutors that scaffold learning through prompt-level guardrails restricting direct answer provision and encouraging hint-based assistance [4].

These scaffolding approaches, while empirically validated, typically apply a fixed instructional strategy across learners or target only one learner stage (e.g., novices [32]), without personalizing support to learners' current stages of skill acquisition. A recent study by Ocheja et al. proposes an LLM-based tutoring system that infers learners' knowledge states from interaction traces using knowledge graphs and adapts feedback accordingly [17]. Cohn et al. similarly introduce a theory-driven architecture for adaptive scaffolding. Their system integrates social-cognitive constructs such as the zone of proximal development and self-efficacy in an LLM-based pedagogical agent [10]. However, despite sophisticated system architectures, both approaches provide limited evidence on how stage-aware scaffolding affects learner engagement and learning processes in an authentic classroom environment.

2.4 Research Questions

While having established that designing tutoring systems that scaffold learners is a promising instructional approach, it remains unclear whether and how learners will actually use such systems. In pluralistic digital learning environments, learners retain substantial freedom to choose whether and which LLM-based tutors to engage with, including instructor-provided tools as well as readily available

general-purpose chatbots. In our setting, the AI tutor is introduced by the course instructor as a chatbot trained on course materials and is embedded directly into the learning management system. This raises a question on whether learners will engage with such a tool when its use is voluntary.

A second concern relates to learners' responses to effort-inducing instructional prompts. When encouraged to self-explain or reason through problems, learners may instead switch to easily accessible, answer-providing alternatives such as ChatGPT. Prior research suggests that learners may prioritize immediacy and convenience over structured instructional support, particularly when low-effort options are available [7, 14]. At the same time, prompts encouraging active processing and self-explanation can foster deeper engagement and learning, provided learners choose to follow them [9]. Thus, the behavioral response to such prompts remains empirically underexplored.

In this study, we are able to observe both decisions simultaneously: whether learners will generally engage with an instructor-endorsed AI tutor in an authentic classroom setting, and whether engagement differs across two versions of the tool due to their instructional design features. We therefore pose the following exploratory research question.

Research Question 1: *How do learners use LLM-based tutors in an authentic classroom setting?*

When implemented appropriately, stage-aware scaffolding is theorized to foster deeper learning activities, such as principle-based reasoning and self-explanation, particularly during the intermediate phase of cognitive skill acquisition [21]. Accordingly, an LLM-based tutor designed to prompt principle recall in early-intermediate learners and to encourage independent problem solving in late-intermediate learners should reduce shallow help-seeking and promote more effortful, reflective engagement. This shift is expected to manifest in more reasoning-oriented dialogue turns and higher mental effort [9]. As highlighted before, there remains scarce causal evidence on how such stage-aware customization in LLM-based tutors will affect learners' reasoning behavior and mental effort in real educational contexts. Thus, we state the second research question.

Research Question 2: *Does learner engagement with stage-aware LLM-based tutors lead to deeper reasoning processes and increased mental effort?*

3 Methodology

3.1 AI Tutor Design

We customized the GPT-4o model [12] using a two-layer architecture (Fig. 1). In the first layer, to improve response accuracy, we integrated the model with course materials (e.g., lecture slides, exercises) via a Retrieval-Augmented Generation

AI Tutor design

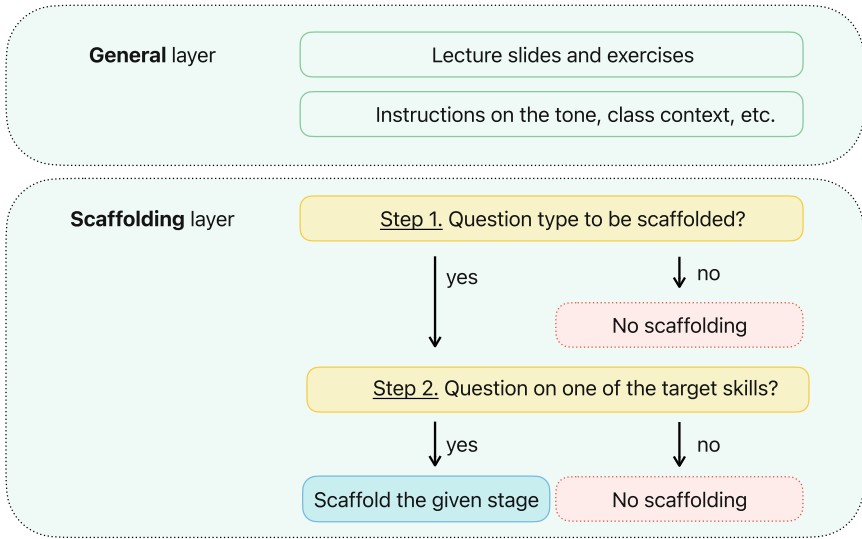


Fig. 1. The design of learner-stage-aware AI tutor.

(RAG) pipeline. Additionally, we augmented the model with context about the course, cohort characteristics, general instructions on tone, etc.

The stage-aware AI tutor included an additional layer, containing instructional logic for scaffolding. This layer required the model to make a two-step decision for each learner question.

1. Determine if the question type warrants scaffolding. When the question was procedural (e.g., how to solve a problem or debug an error) or conceptual (e.g., explain a topic), the tutor was prompted to consider scaffolding. For factual or unrelated questions, the tutor responded as usual without scaffolding.
2. Check alignment with course core skills. When the question was eligible for scaffolding, the tutor next checked whether it related to one of ten core skills predefined by the instructors (e.g., selection, aggregation, basic plotting). If so, scaffolding was applied.

When scaffolding was required, the tutor followed additional stage-specific guidance, as outlined in Table 1.

3.2 Field Experiment

Course Setup. We conducted a randomized controlled experiment in a 17-week master's-level programming course to compare a stage-aware LLM-based tutor with a baseline LLM-based tutor. The baseline tutor included the same General layer but not the Scaffolding layer (Fig. 1).

Table 1. Stages, learner states and scaffolding instructions for the meta-prompt.

Stage	Reference name	Learner state	Instructions
Stage 1	Novice	Reliance on worked examples, limited schema formation, high cognitive load	Provide worked-out examples, make solution rationale explicit, minimize problem-solving demands
Stage 2	Early-intermediate	Self-explanations, principle recall, partial solution completion	Remind principles, prompt self-explanation and first steps to solution
Stage 3	Late-intermediate	Independent problem solving, schema refinement, reduced need for guidance	Fade scaffolding, prompt full solution attempts, provide corrective feedback only when needed

The course followed a flipped format: students engaged in self-study of the materials and completed three analytics projects: the first two on-site, the third online in asynchronous format. Assessment consisted of a final exam, graded assessments following each in-class analytics project and an evaluation of the third analytics project.

The experimental phase took place during the first five weeks of the course. During this period, students learned introductory programming concepts through self-study and participated in a 3.5-hour in-person analytics project, where they applied these concepts to a practical problem. Immediately following this session, students completed a graded multiple-choice assessment involving coding tasks. Participants had access to their assigned AI tutor during both the self-study phase and the in-class analytics project, but not during the graded assessment.

Experimental Setup. All students enrolled in the course ($N = 28$) were invited and agreed to participate in the study. Participation was incentivized through a small bonus added to the final course grade, as well as a small donation made to an NGO for each participant.² At the beginning of the course, students reported their prior (baseline) programming skills in the chosen language (R or Python), which we mapped to learning stages as follows: little or no prior exposure to programming (novice, $N = 17$), only basic knowledge of programming (early-intermediate, $N = 9$), and prior experience with larger programming assignments (late-intermediate, $N = 2$). They were then randomly assigned to one of the two

² The study was approved by the faculty’s ethics review board at the University of Zürich.

tutor conditions, with stratification based on their baseline skill level. At the end of the experiment students completed an endline survey.

We measured three outcomes: (1) tutor usage, based on system logs collected during the experiment ($N = 19$); (2) mental effort, assessed in the endline survey ($N = 20$); and (3) performance on the graded assignment, scored from 0 to 6 based on the number of correctly answered questions ($N = 26$). Differences in sample size reflect that not all students used the tutor, completed the endline survey or participated in the graded assignment.

4 Results

4.1 AI Tutor Validation

To evaluate whether the customized AI tutor responds as instructed, we examined student–tutor interactions to assess the quality of the tutor’s responses. For each learner stage, we first determined, based on the question type and the referenced skill, whether the question *should have been* scaffolded according to the design guidelines. Next, for questions that *were in fact* scaffolded, we assessed whether the tutor’s response followed the stage-appropriate scaffolding instructions. We measure accuracy as the proportion of correct decisions.

As shown in Table 2, the AI tutor followed stage-specific instructional guidelines (denoted as *Appropriate instruction* in the table) with near-perfect fidelity once scaffolding was triggered. The errors primarily arose in the preceding decision of whether to scaffold (denoted as *Scaffolding event* in the table), particularly for late-intermediate learners, where the tutor occasionally provided direct assistance instead of prompting further reasoning (accuracy = 0.84). This estimate however is based on a small number of messages in the group.

To verify the validity of these results, one author annotated the full dataset, and another author independently annotated a subset of the first interactions in randomly selected 20% of conversations. Percent agreement was 74% for judgments of whether scaffolding was warranted and 68% for detecting whether scaffolding occurred; agreement for assessing the appropriateness of scaffolded responses was 100%. These results highlight that deciding whether scaffolding is warranted is non-trivial even for human annotators. At the same time, once scaffolding occurred, its stage-specific appropriateness was unambiguous.

4.2 RQ1. How Do Learners Use LLM-Based Tutors in an Authentic Classroom Setting?

We first examined overall usage of the AI tutor. Although students were encouraged to use the tutor, engagement was entirely voluntary, providing a naturalistic setting for observing usage patterns. Analysis of the system logs ($N = 759$ learner messages) showed that 19 of the 28 students (68%) used the tutor at least once, and that 14 students (50%) engaged with the tutor on at least two separate days. Usage increased steadily over the experimental period, peaking on the final day during the analytics project, when 46% of all student messages were sent.

Table 2. AI tutor scaffolding verification.

Task	Level	N messages	Accuracy
Scaffolding event	Novice	221	0.95
	Early-Intermediate	199	0.94
	Late-Intermediate	31	0.84
Appropriate instruction	Novice	180	1.00
	Early-Intermediate	107	0.97
	Late-Intermediate	15	1.00

To better understand how the tutor was used, we manually annotated all messages to: (1) group messages belonging to the same conversation; and (2) identify the associated learning activity (same for all messages in the conversation: practice exercises during self-study, ungraded quizzes during self-study, in-class analytics project, other course-related activity, or unrelated to the course). Most messages were related to solving the exercises in the self-study period (57%), followed by preparing for the analytics project (32%), other content acquisition questions (6%) and content-unrelated questions (3%) and quizzes (2%). Figure 2 summarizes these results.

Next, we examined individual *usage patterns and compared them across the two groups* (Fig. 3). To compare group means, we conducted one-sided Bayesian independent-samples t-tests [27] with a Cauchy prior ($r = 0.707$) on the standardized effect size. We tested the directional hypothesis that the stage-aware AI tutor increases usage relative to the control condition. The Bayes factor (BF_{+0}) quantifies the evidence for this directional alternative relative to the null hypothesis of no difference (i.e., how much more likely the data is under the alternative than under the null); values greater than 1 indicate evidence in favor of higher usage in the stage-aware condition. Conversely, BF_{-0} quantifies evidence for the hypothesis that the stage-aware tutor reduces usage relative to the null; values less than 1 therefore indicate evidence against a reduction in usage. Students who received the stage-aware AI tutor wrote more messages on average ($M_{treat} = 56.38$, $SD = 47.63$, $n = 8$, $M_{control} = 28.00$, $SD = 25.50$, $n = 11$, $BF_{+0} = 1.90$, $BF_{-0} = 0.19$), and used the tutor on more days ($M_{treat} = 4.25$, $SD = 2.76$, $n = 8$, $M_{control} = 2.27$, $SD = 1.35$, $n = 11$, $BF_{+0} = 3.10$, $BF_{-0} = 0.17$.) Overall, the findings indicate evidence for increased usage (and against reduction in usage) in the stage-aware condition.

We complemented the usage log analysis with data from the endline survey. When asked about the benefits of the AI tutor, students most frequently highlighted its adaptation to the course materials (40%) and the clarity of its explanations (25%). Regarding reasons for not using the tutor, or for switching to alternative tools such as ChatGPT, 15% of respondents reported the inability to upload images as a limitation. While image-based input may appear convenient, prior research suggests that the process of explicitly formulating questions and

representing problems is itself cognitively demanding and beneficial for learning [26].

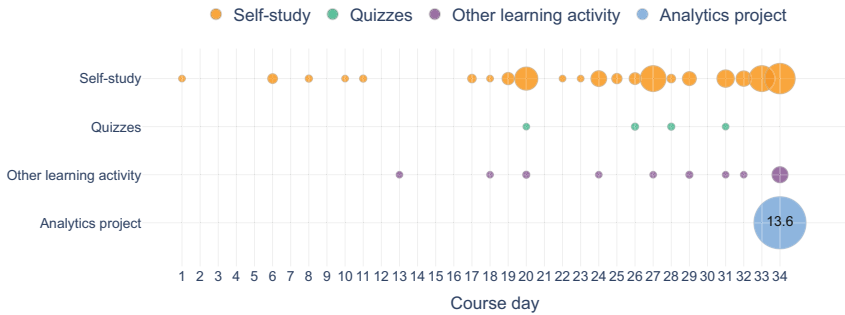


Fig. 2. Daily use of the AI tutor across different learning activities. The y-axis represents learning activities. The x-axis represents days of the course. Each circle represents the average number of messages sent by students on a given day. Circles where $N > 10$ are labeled with a corresponding average.

4.3 RQ2. Does Learner Engagement with Stage-Aware LLM-Based Tutors Lead to Deeper Reasoning Processes and Increased Mental Effort?

We examined the types of interactions underlying the observed differences in engagement. One author manually annotated each course-related message to classify the self-regulated learning (SRL) strategy reflected in the message (help-seeking, effort regulation, elaboration, critical thinking, organization and rehearsal) following an established framework [18–20]. Another author independently annotated first user messages in the 20% of the conversations, yielding an inter-rater agreement of 74%. Most messages reflect help-seeking (54%), followed by elaboration (24%), critical thinking (16%), effort regulation (2%), and organization (1%). As expected from literature documenting learning strategies in similar setups [8, 30], we do not observe any rehearsal questions. Finally, some messages were unrelated to the SRL scheme, such as clarification questions (3%). We further considered only messages that are both course and SRL scheme related ($N=713$). Given the low counts in several categories, we grouped messages into *deep* (elaboration, critical thinking, effort regulation) and *shallow* (help-seeking, organization) interactions. Averaged across all messages, the proportions of deep interactions was relatively comparable across treatment groups ($M_{treat} = 44\%$, $M_{control} = 42\%$). To examine whether the AI tutor influenced the depth of follow-up interactions, we focused on the second message in each conversation, as it is not directly tutor-induced. For each learner, we computed the proportion of initial messages that were followed by a deep message (as opposed

to a shallow or no follow-up). These learner-level proportions were compared between conditions using the same one-sided Bayesian independent-samples t-test described above. Results indicated that learners in the stage-aware condition followed up with deep messages more frequently than those in the control condition ($M_{treat} = 0.28$, $SD = 0.23$, $n = 8$, $M_{control} = 0.14$, $SD = 0.13$, $n = 10$, $BF_{+0} = 1.89$, $BF_{-0} = 0.19$.)³ Overall, the results provide initial evidence that the learner-stage-aware LLM-based tutors lead to deeper reasoning processes.

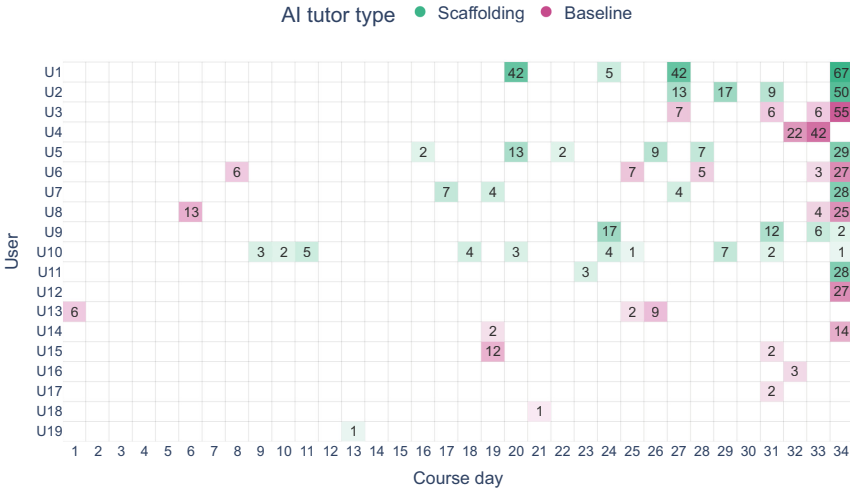


Fig. 3. Daily use of the AI tutor across experimental conditions. The y-axis represents different users. The x-axis represents days of the course. Each cell represents the number of messages sent by a given user on a given day. Darker green and darker purple represent increased usage. Users are sorted according to their activity.

Mental Effort. We measured self-reported mental effort in the endline survey as the mean of the four items on the 5-point Likert scale presented in [11]. On average, students in the stage-aware tutor condition reported slightly higher mental effort than those in the control condition ($M_{treat} = 4.3$, $SD = 0.45$, $n = 10$, $M_{control} = 3.9$, $SD = 0.77$, $n = 10$, $BF_{+0} = 1.37$, $BF_{-0} = 0.20$.), indicating slightly higher cognitive engagement during interactions with the tutor.

This effect was more pronounced for the subgroup of learners at early- and late-intermediate skill levels (as opposed to novice, ($M_{treat} = 4.42$, $SD = 0.52$, $n = 3$, $M_{control} = 3.5$, $SD = 0.66$, $n = 3$, $BF_{+0} = 2.07$, $BF_{-0} = 0.31$.) This pattern is theoretically consistent with prior work suggesting that, beyond the novice stage, instructional prompts that foster reasoning, self-explanation, and

³ One learner in the control condition wrote only one question (unrelated to course) and was thus excluded from this analysis.

problem decomposition tend to increase invested mental effort while supporting deeper learning processes [25].

Lastly, we examined the effect of the stage-aware AI tutor on academic performance, measured by the evaluation score in the analytics project. Mean grades were comparable between the treatment and control groups ($M_{treat} = 4.85$, $SD = 0.90$, $n = 13$, $M_{control} = 4.92$, $SD = 0.76$, $n = 13$, $BF_{+0} = 0.31$, $BF_{-0} = 0.43$). The Bayes factors indicate that the data provide more evidence for the null hypothesis of no performance difference than for either directional alternative. One possible explanation is limited measurement sensitivity, as the evaluation consisted of only six questions, which may have constrained its ability to detect group differences. A second explanation is that evaluation format (an open-book, multiple-choice test) does not allow to disentangle independent problem-solving from efficient solution retrieval.

5 Discussion and Conclusion

This work introduces a stage-aware LLM-based tutor and provides initial empirical evidence on how it is used by learners in an authentic classroom setting, and whether stage-level customization promotes deeper reasoning and greater mental effort.

First, the results indicate that learners are generally willing to engage with a course-embedded AI tutor when its use is voluntary, although frequent use (on more than two days) remains moderate. One system-specific explanation is the absence of image-based input, a feature often associated with convenience in general-purpose LLM tools, despite limited evidence of its effectiveness for learning. More broadly, this suggests that learner engagement is shaped not only by instructional quality of an AI tutor but also by competition with conventional AI systems that might prioritize immediacy and low effort. Future research should therefore disentangle the mechanisms underlying learners' preferences across AI tools, particularly how specific design features influence adoption, abandonment, and learning outcomes.

Second, our findings provide initial evidence that stage-aware LLM-based tutoring fosters deeper cognitive engagement, reflected in both higher depth of learner–tutor interactions and higher reported mental effort. Consistent with the tutor's design, which prompts learners in the early and late intermediate stages to think, we observe that learners respond positively to stage-aware scaffolding: rather than disengaging when prompted to reflect or explain, they tend to remain engaged and produce reasoning-oriented responses. This pattern is consistent with theories of cognitive skill acquisition, which suggest that prompting principle recall and partial solution generation is most effective once foundational schemas are established. The observed increase in self-reported mental effort, particularly among intermediate learners, further indicates that stage-aware scaffolding can elicit more cognitively demanding and potentially more productive engagement.

Two important limitations of the study are the small sample size and the relatively short duration of the experimental period. Extending the implementation to larger courses and to longer instructional periods will help validate the generalizability of the findings.

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Reproducibility. Code used for data analysis is available in the following repository: <https://github.com/mpoiaganova/stage-aware-AI-tutor>. This repository additionally provides customization meta-prompts and further details on the AI tutor setup. Anonymized user data are available upon request; message content may be removed to protect privacy.

References

1. Akhuseyinoglu, K., et al.: A study of worked examples for SQL programming. In: Proceedings of the 27th ACM Conference on Innovation and Technology in Computer Science Education, vol. 1, pp. 82–88 (2022)
2. Anderson, J.R.: Acquisition of cognitive skill. *Psychol. Rev.* **89**(4), 369 (1982)
3. Atkinson, R.K., Derry, S.J., Renkl, A., Wortham, D.: Learning from examples: instructional principles from the worked examples research. *Rev. Educ. Res.* **70**(2), 181–214 (2000)
4. Bastani, H., Bastani, O., Sungu, A., Ge, H., Kabakcı, Ö., Mariman, R.: Generative AI without guardrails can harm learning: evidence from high school mathematics. *Proc. Natl. Acad. Sci.* **122**(26), e2422633122 (2025)
5. Bauer, E., Greiff, S., Graesser, A.C., Scheiter, K., Sailer, M.: Looking beyond the hype: understanding the effects of AI on learning. *Educ. Psychol. Rev.* **37**(2), 45 (2025)
6. Bhaskar, S., Hamouda, S.: Designing answer-aware LLM hints to scaffold deeper learning in k-12 programming education. In: Proceedings of the 2025 ACM Conference on International Computing Education Research, vol. 2, pp. 7–8 (2025)
7. Borchers, C., Fleischer, H., Schanze, S., Scheiter, K., Aleven, V.: High scaffolding of an unfamiliar strategy improves conceptual learning but reduces enjoyment compared to low scaffolding and strategy freedom. *Comput. Educ.* 105364 (2025)
8. Cheng, G., Zou, D., Xie, H., Wang, F.L.: Exploring differences in self-regulated learning strategy use between high-and low-performing students in introductory programming: an analysis of eye-tracking and retrospective think-aloud data from program comprehension. *Comput. Educ.* **208**, 104948 (2024)
9. Chi, M.T., Bassok, M., Lewis, M.W., Reimann, P., Glaser, R.: Self-explanations: how students study and use examples in learning to solve problems. *Cogn. Sci.* **13**(2), 145–182 (1989)

10. Cohn, C., et al.: A theory of adaptive scaffolding for LLM-based pedagogical agents. arXiv preprint [arXiv:2508.01503](https://arxiv.org/abs/2508.01503) (2025)
11. Endres, T., et al.: Developing the mental effort and load-translingual scale (MELTS) as a foundation for translingual research in self-regulated learning. *Educ. Psychol. Rev.* **37**(1), 5 (2025)
12. Hurst, A., et al.: GPT-4o system card. arXiv preprint [arXiv:2410.21276](https://arxiv.org/abs/2410.21276) (2024)
13. Kazemitabaar, M., Huang, O., Suh, S., Henley, A.Z., Grossman, T.: Exploring the design space of cognitive engagement techniques with AI-generated code for enhanced learning. In: *Proceedings of the 30th International Conference on Intelligent User Interfaces*, pp. 695–714 (2025)
14. Koedinger, K.R., Aleven, V.: Exploring the assistance dilemma in experiments with cognitive tutors. *Educ. Psychol. Rev.* **19**(3), 239–264 (2007)
15. Ma, S., Wang, J., Zhang, Y., Ma, X., Wang, A.Y.: DBox: scaffolding algorithmic programming learning through learner-LLM co-decomposition. In: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pp. 1–20 (2025)
16. McLaren, B.M., Van Gog, T., Ganoë, C., Karabinos, M., Yaron, D.: The efficiency of worked examples compared to erroneous examples, tutored problem solving, and problem solving in computer-based learning environments. *Comput. Hum. Behav.* **55**, 87–99 (2016)
17. Ocheja, P., Flanagan, B., Dai, Y., Ogata, H.: How good is ChatGPT in giving adaptive guidance using knowledge graphs in e-learning environments? arXiv preprint [arXiv:2412.03856](https://arxiv.org/abs/2412.03856) (2024)
18. Pintrich, P.R.: The role of motivation in promoting and sustaining self-regulated learning. *Int. J. Educ. Res.* **31**(6), 459–470 (1999)
19. Pintrich, P.R.: A conceptual framework for assessing motivation and self-regulated learning in college students. *Educ. Psychol. Rev.* **16**(4), 385–407 (2004)
20. Pintrich, P.R., Smith, D.A., Garcia, T., McKeachie, W.J.: Reliability and predictive validity of the motivated strategies for learning questionnaire (MSLQ). *Educ. Psychol. Measur.* **53**(3), 801–813 (1993)
21. Renkl, A.: Learning from worked-out examples: a study on individual differences. *Cogn. Sci.* **21**(1), 1–29 (1997)
22. Renkl, A.: Toward an instructionally oriented theory of example-based learning. *Cogn. Sci.* **38**(1), 1–37 (2014)
23. Renkl, A., Atkinson, R.K.: Structuring the transition from example study to problem solving in cognitive skill acquisition: a cognitive load perspective. In: *Cognitive Load Theory*, pp. 15–22. Routledge (2016)
24. Renkl, A., Atkinson, R.K., Große, C.S.: How fading worked solution steps works—a cognitive load perspective. *Instr. Sci.* **32**(1), 59–82 (2004)
25. Renkl, A., Atkinson, R.K., Maier, U.H., Staley, R.: From example study to problem solving: smooth transitions help learning. *J. Exp. Educ.* **70**(4), 293–315 (2002)
26. Rosenshine, B., Meister, C., Chapman, S.: Teaching students to generate questions: a review of the intervention studies. *Rev. Educ. Res.* **66**(2), 181–221 (1996)
27. Rouders, J.N., Speckman, P.L., Sun, D., Morey, R.D., Iverson, G.: Bayesian t tests for accepting and rejecting the null hypothesis. *Psychon. Bull. Rev.* **16**(2), 225–237 (2009)
28. Sweller, J., Van Merriënboer, J.J., Paas, F.G.: Cognitive architecture and instructional design. *Educ. Psychol. Rev.* **10**(3), 251–296 (1998)
29. VanLehn, K.: Cognitive skill acquisition. *Annu. Rev. Psychol.* **47**(1), 513–539 (1996)

30. Viberg, O., Wong, J., Feldman-Maggor, Y., Dunder, N., Epp, C.D.: Chatting with code: exploring LLMs as learning partners in programming education. In: International Conference on Artificial Intelligence in Education, pp. 453–461. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-98465-5_57
31. Waldeyer, J., Endres, T., Roelle, J., Baars, M., Renkl, A.: How to optimize self-assessment accuracy in cognitive skill acquisition when learning from worked examples. *Educ. Psychol. Rev.* **36**(4), 103 (2024)
32. Wang, W., et al.: Novices' learning barriers when using code examples in open-ended programming. In: Proceedings of the 26th ACM Conference on Innovation and Technology in Computer Science Education, vol. 1, pp. 394–400 (2021)
33. Xie, W., Shi, H., Li, Y., Liu, J.: STAP: a socratic tutor for adaptive programming with pedagogical scaffolding. In: Proceedings of the 2025 2nd International Symposium on Artificial Intelligence for Education, pp. 584–587 (2025)